



Internship Presentation

on

Analysis and Implementation of FFT Algorithms

by

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Internship carried out at

Advanced Systems Laboratory

Defence Research And Development Organisation

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1. About The Organisation: DRDO

- Premier agency under the Ministry of Defence, estd: 1958
- Responsible for the research and development of defence technologies.
- Has network of over 50 laboratories, specializing in various disciplines such as aeronautics, armaments, electronics, missiles, combat vehicles, naval systems, advanced computing, simulation, and life sciences.
- **Advanced System Laboratory (ASL)**: engaged in the development of solid propulsion technologies, composites, aerospace mechanisms, etc



2. Overview of internship

- Fourier Transform is computationally expensive
- Internship involved understanding, analyzing and implementing few FFT algorithms using SystemVerilog in the context of FPGA
- Implemented Radix-2 Inplace, R2MDC, R2SDC and Winograd
- For resource analysis Artix-7 (xc7a100tcsg324-1) family of FPGA was targeted
- Specs:
 - 15,850 CLB slices (4x6-input LUT, 8 FF)
 - 4860 Kb dual-port block RAM
 - 240 DSP slices with 25 x 18 multiplier, 48-bit accumulator
 - 101,440 Logic cells
 - 13 Mb Block RAM
 - 300 User I/O

3. Techniques

3.1 Introduction

- Implemented Radix-2 Inplace, R2MDC, R2SDC and 5 point Winograd
- Transform size is parameterized in Radix-2 Inplace, R2MDC, R2SDC
- Winograd is not scalable: needs different implementation for different transform sizes

3.2 Method used in analysis

- Simulation and debugging was done using Verilator
- Synthesis and implementation in Vivado
- For automating report generation in Vivado, `tcl` script was used
- Assumptions:
 - 10ns clock
 - control signal toggle rate: 2%
 - toggle rate & static probability for I/O: 50%
- Throughput :

$$\frac{\text{Number of samples}}{\text{Number of cycles to process whole input}} \times \text{max frequency of operation}$$

3.3 Cooley–Tukey: Radix-2 DIT

- CT: Re-expresses the DFT of an size $N_1 N_2$ in terms of N_1 smaller DFTs of sizes N_2
- Specific methods: Radix-2, Radix-4, mixed-radix
- For Radix-2 overall time complexity : $\mathcal{O}(N \log N)$

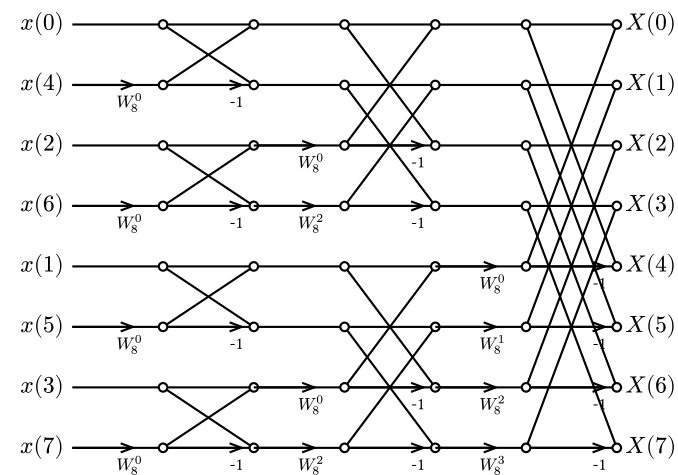


Figure 1: Signal flow graph of 8-point radix-2 DIT-FFT

3.3 Cooley–Tukey: Radix-2 DIT

3.3.1 Implementation

- Burst architecture
- I/O are fixed point complex numbers
- Output available after $N \log_2(N)/2$ cycles after input has been loaded

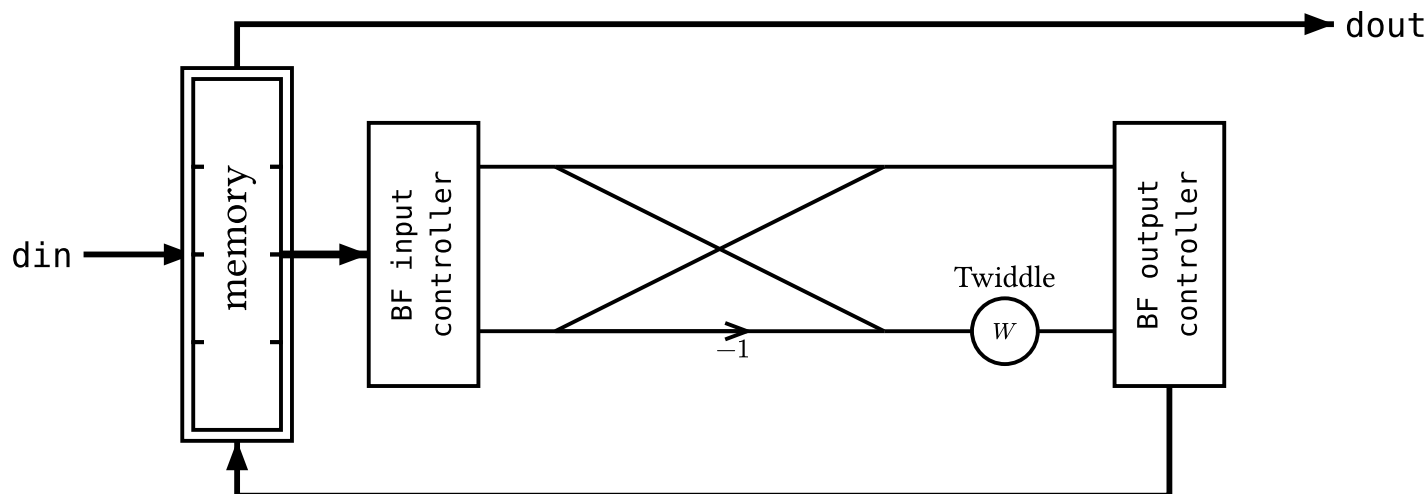


Figure 2: Radix-2 In-place implementation

3.3 Cooley–Tukey: Radix-2 DIT

3.3.2 Simulation result

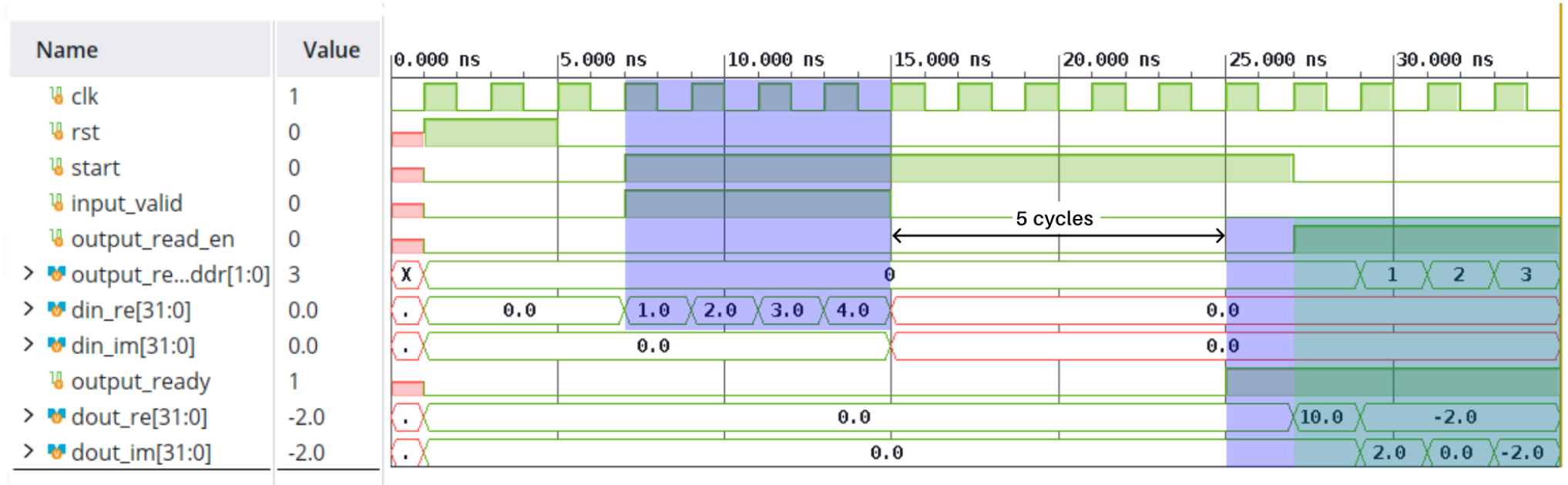


Figure 3: Simulation result of 4 point Radix-2 DIT-FFT for a ramp input

3.4 Radix-2 Multi Path Delay Commutator (R2MDC)

- Based on the Decimation in Frequency (DIF)
- Takes 1 input and produces 2 outputs per cycle

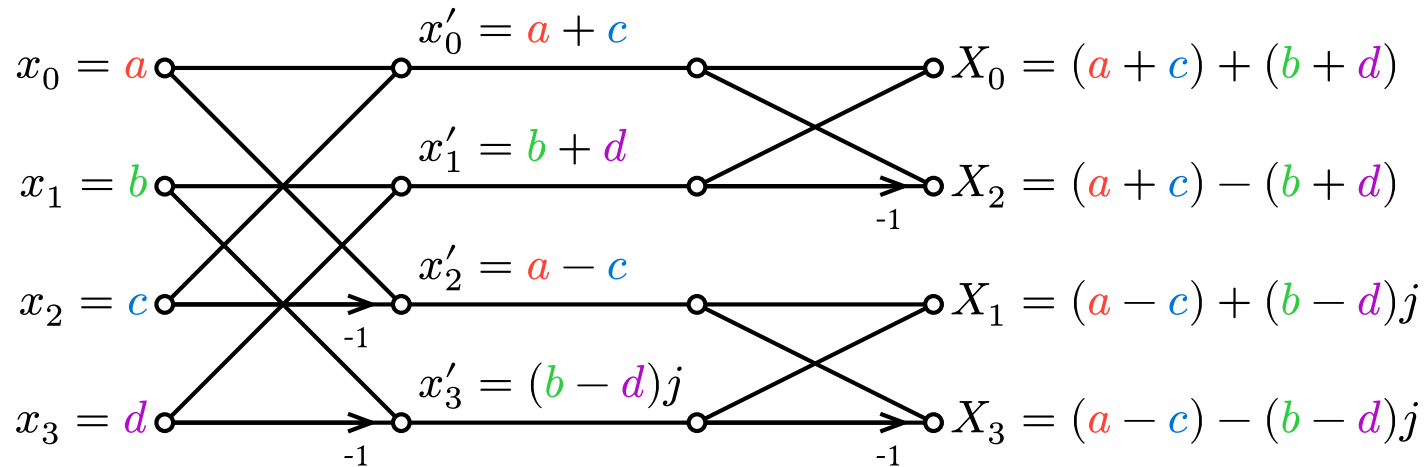


Figure 4: Signal flow graph of 4-point radix-2 DIF-FFT

3.4 Radix-2 Multi Path Delay Commutator (R2MDC)

cycle

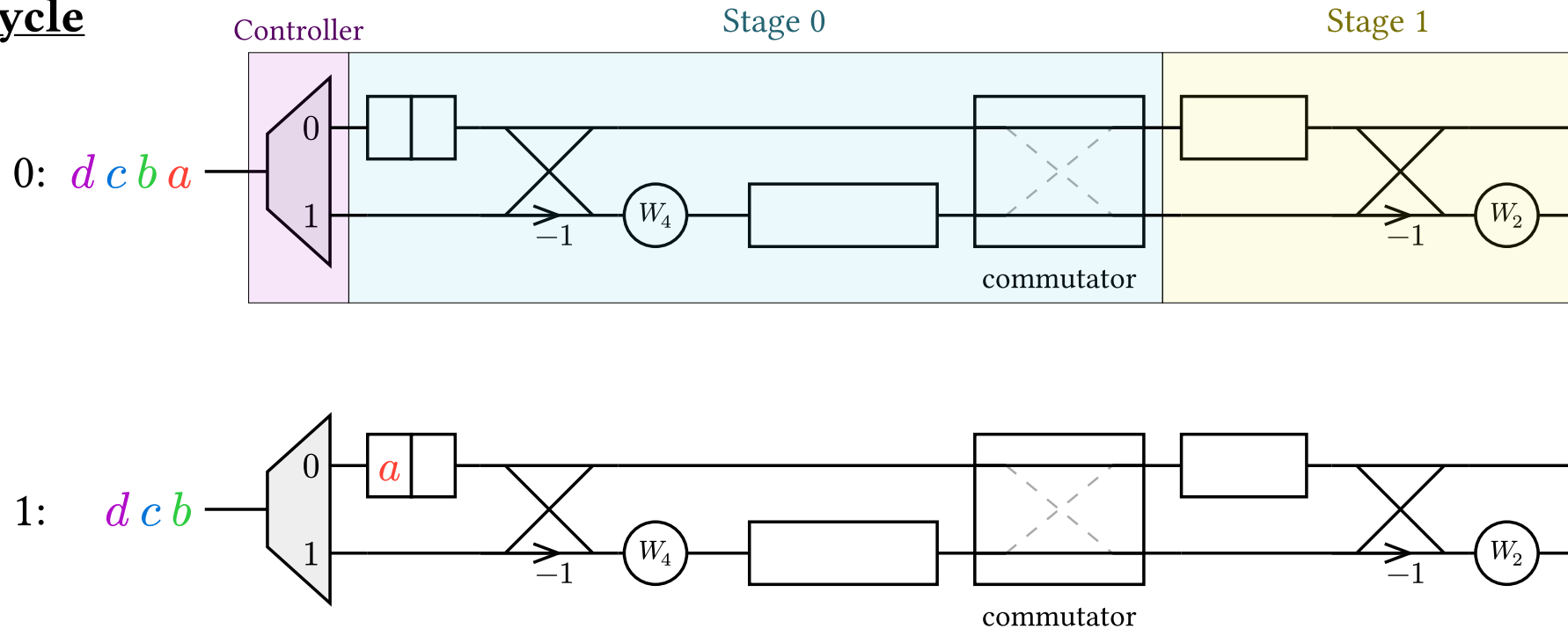


Figure 5: R2MDC algorithm for 4 point case

3.4 Radix-2 Multi Path Delay Commutator (R2MDC)

cycle

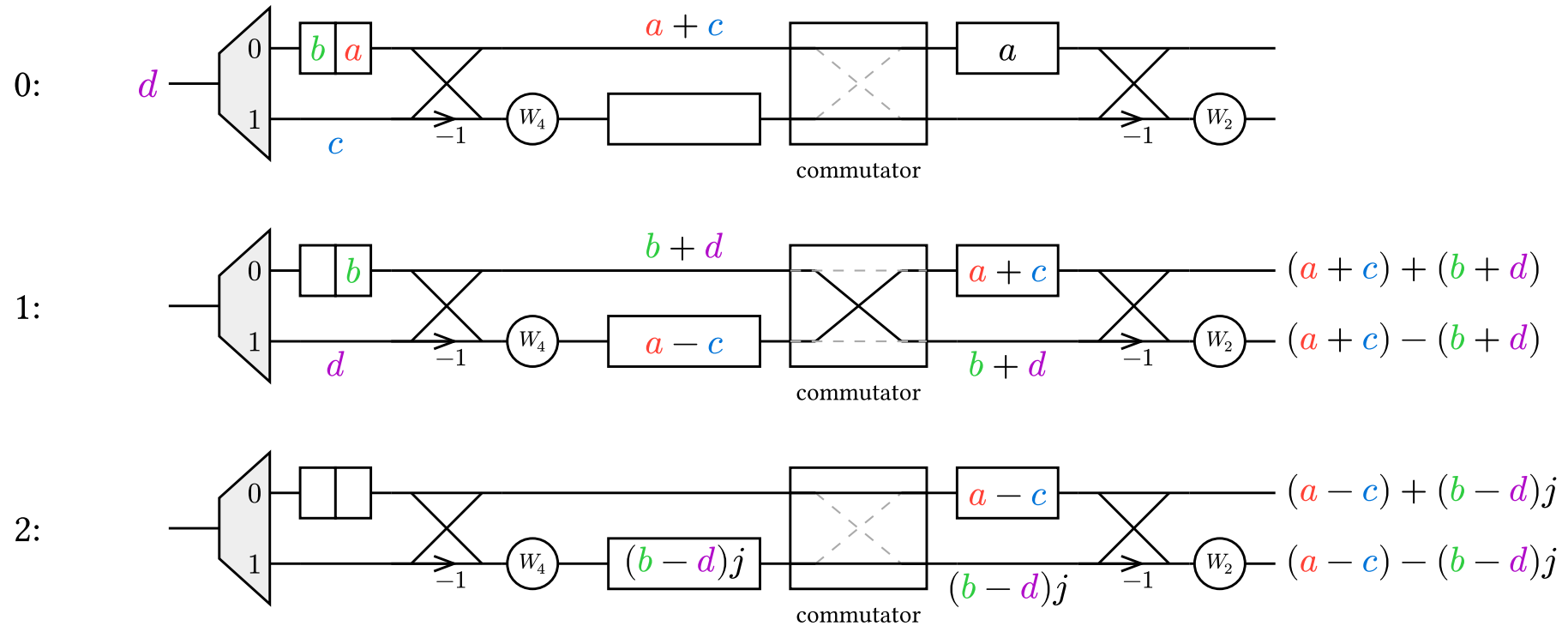


Figure 6: R2MDC algorithm for 4 point case

3.4 Radix-2 Multi Path Delay Commutator (R2MDC)

3.4.1 Implementation

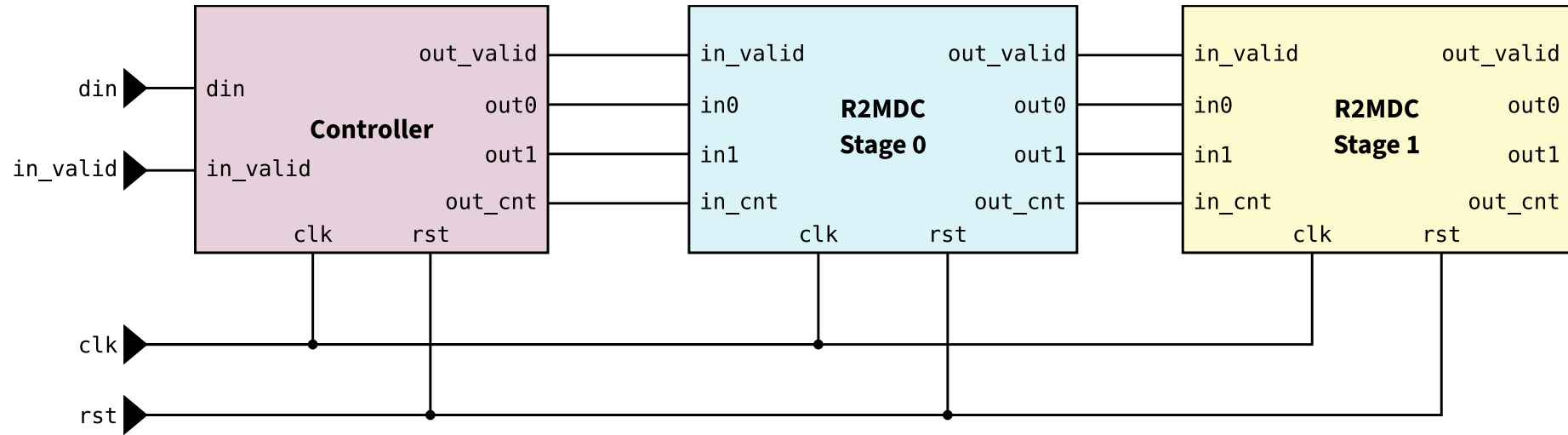


Figure 7: Schematic of R2MDC implementation

3.4 Radix-2 Multi Path Delay Commutator (R2MDC)

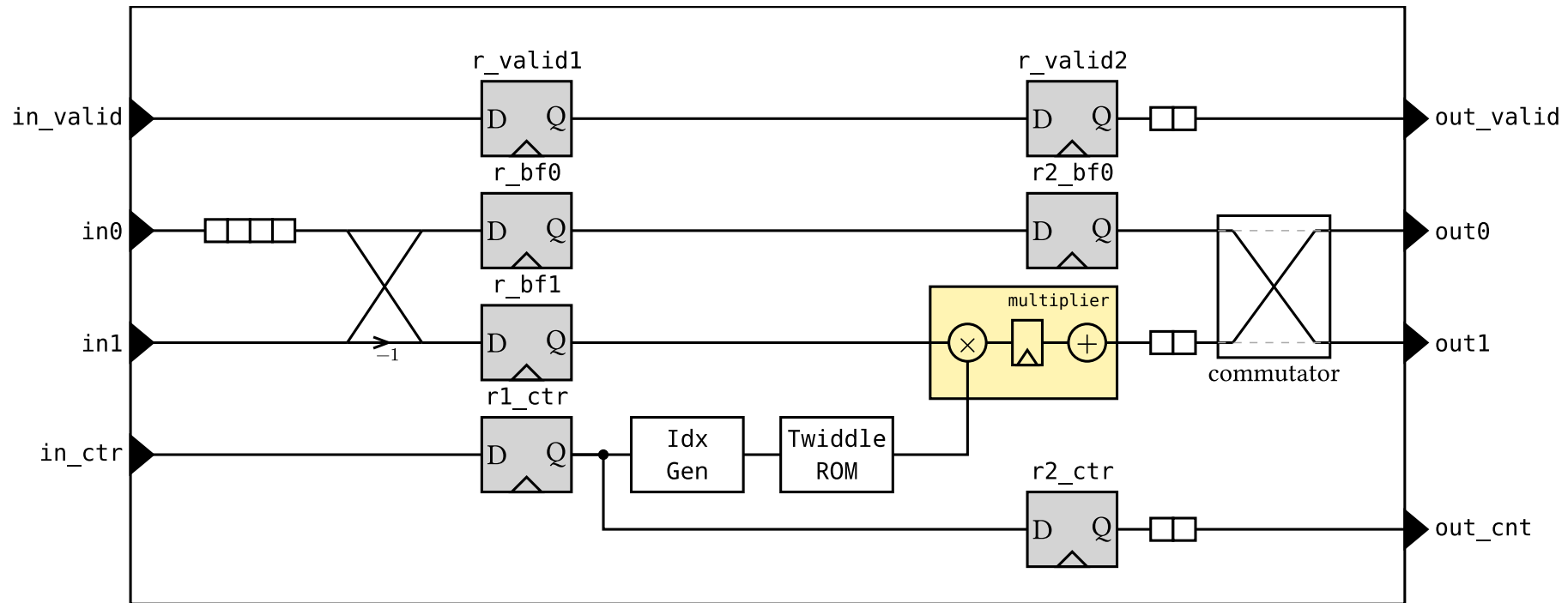


Figure 8: Inside a single R2MDC Stage. All modules with wedge is clocked

3.4 Radix-2 Multi Path Delay Commutator (R2MDC)

3.4.2 Simulation result

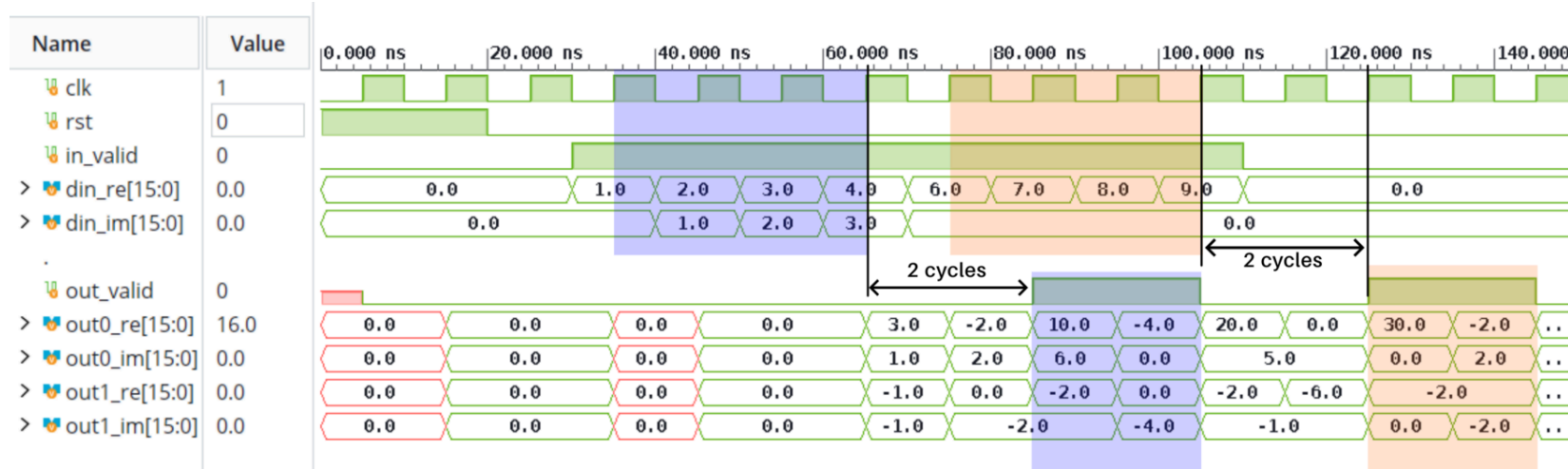


Figure 9: Simulation result for two 4 point test signals in R2MDC

- 2 different colors → different inputs.
- 2 outputs are obtained at same clock cycle
- Takes $\log_2(N)$ cycles to get all outputs after last input has inserted.

3.5 Radix-2 Single Path Delay Commutator (R2SDC)

- Another pipelined implementation of DIF algorithm
- Produces single output per cycle
- Uses fewer number of shift registers compared to MDC (expense of higher latency and complexity)

3.5 Radix-2 Single Path Delay Commutator (R2SDC)

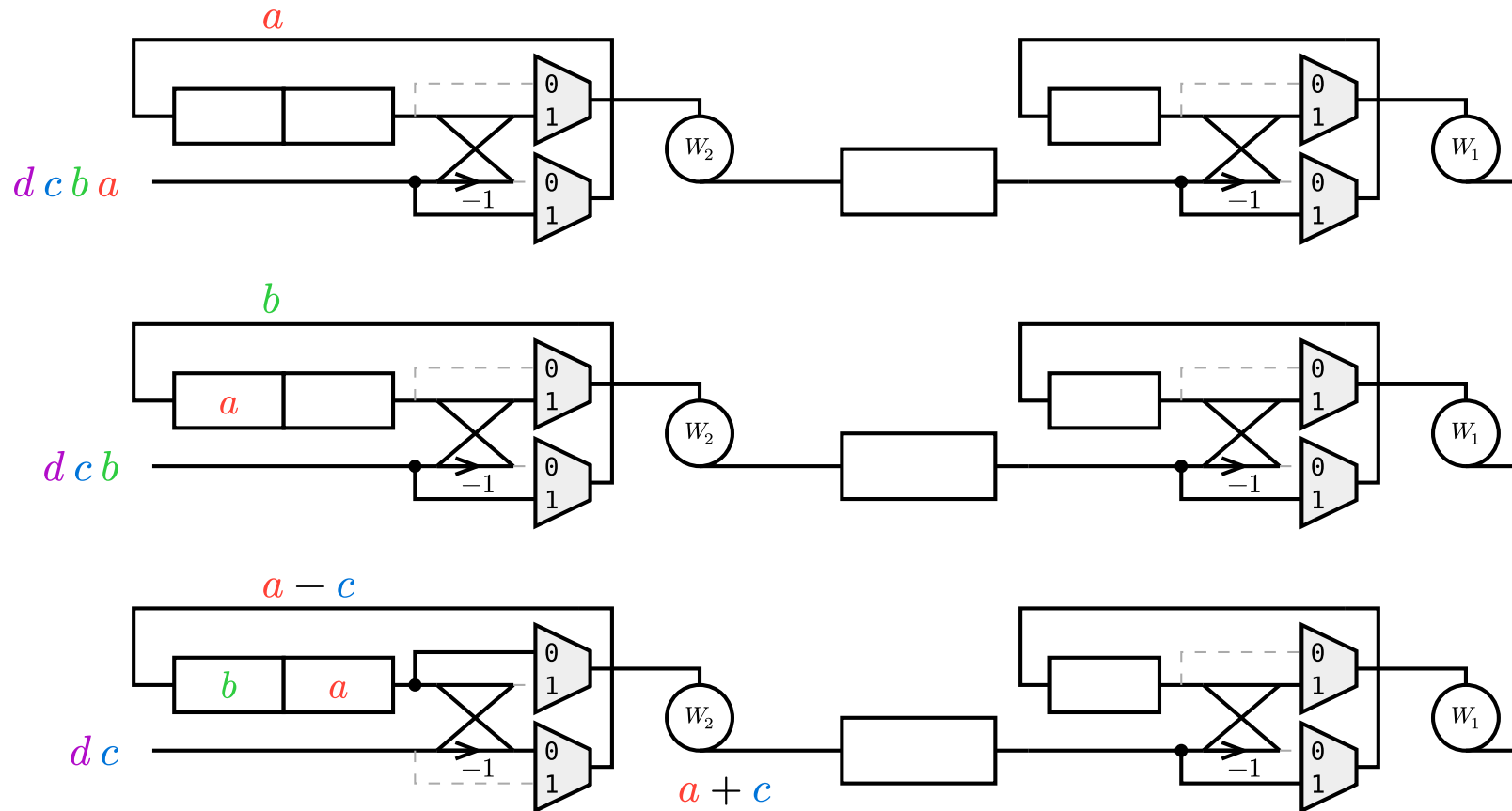


Figure 10: Steps involved in a 4 point R2SDC algorithm

3.5 Radix-2 Single Path Delay Commutator (R2SDC)

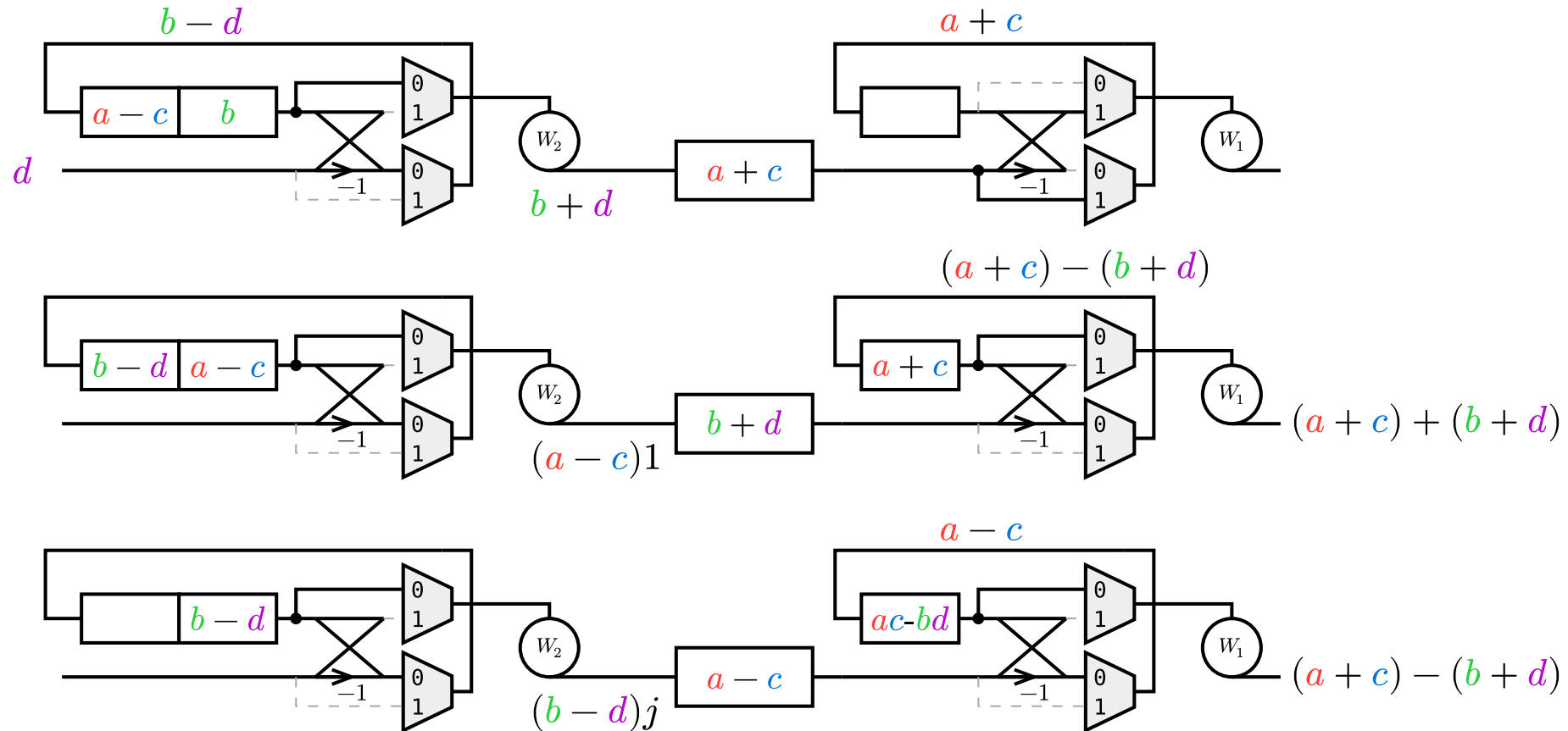


Figure 11: Steps involved in a 4 point R2SDC algorithm

3.5 Radix-2 Single Path Delay Commutator (R2SDC)

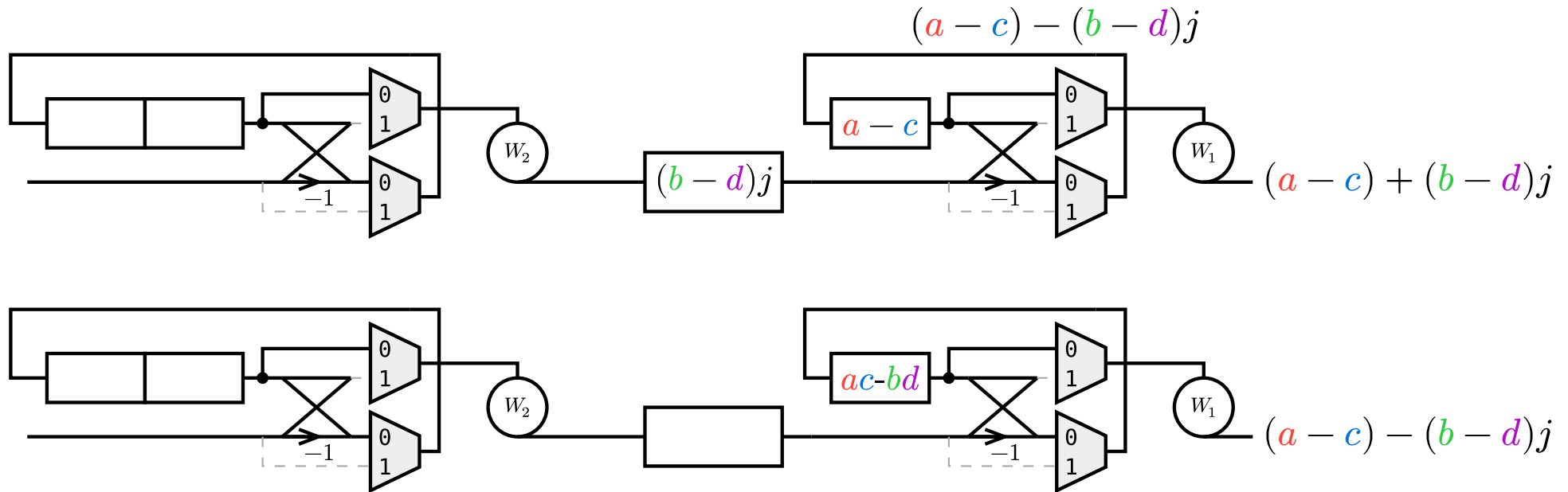


Figure 12: Steps involved in a 4 point R2SDC algorithm

3.5 Radix-2 Single Path Delay Commutator (R2SDC)

3.5.1 Implementation

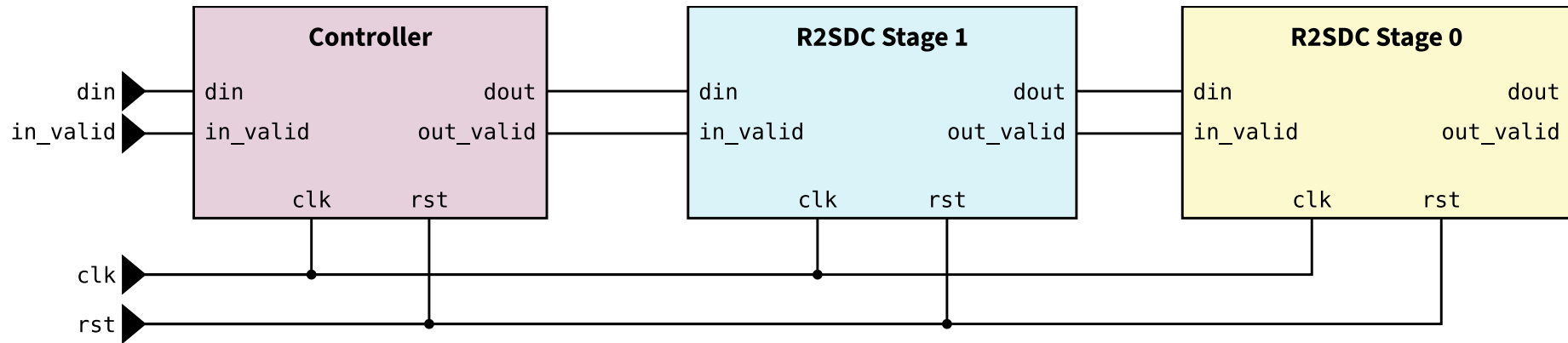


Figure 13: Schematic of R2SDC implementation

3.5 Radix-2 Single Path Delay Commutator (R2SDC)

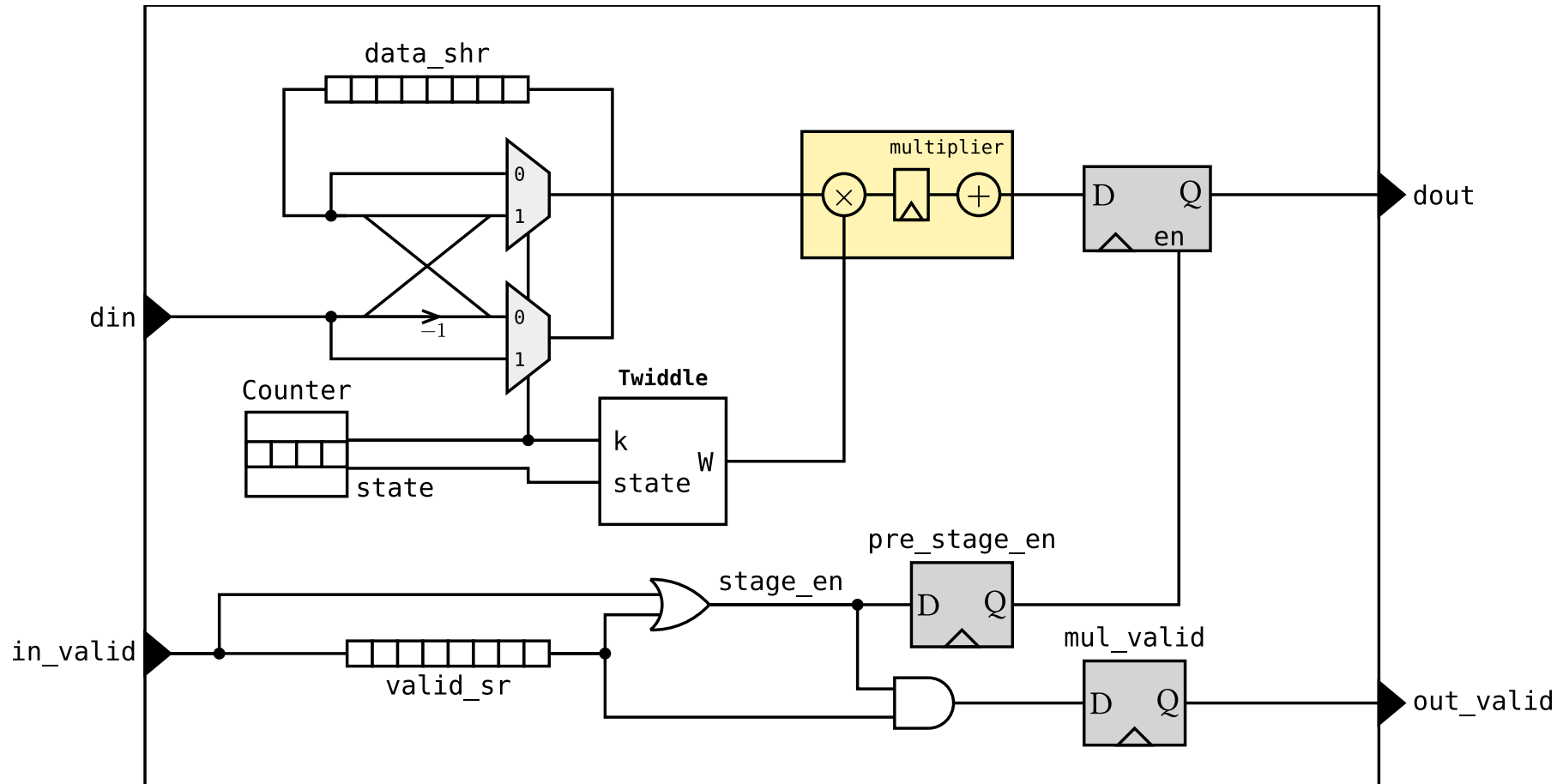


Figure 14: Inside a single R2SDC Stage. All modules with wedge is clocked

3.5 Radix-2 Single Path Delay Commutator (R2SDC)

3.5.2 Simulation result

- The output arrives one clock cycle later.
- Burst architecture

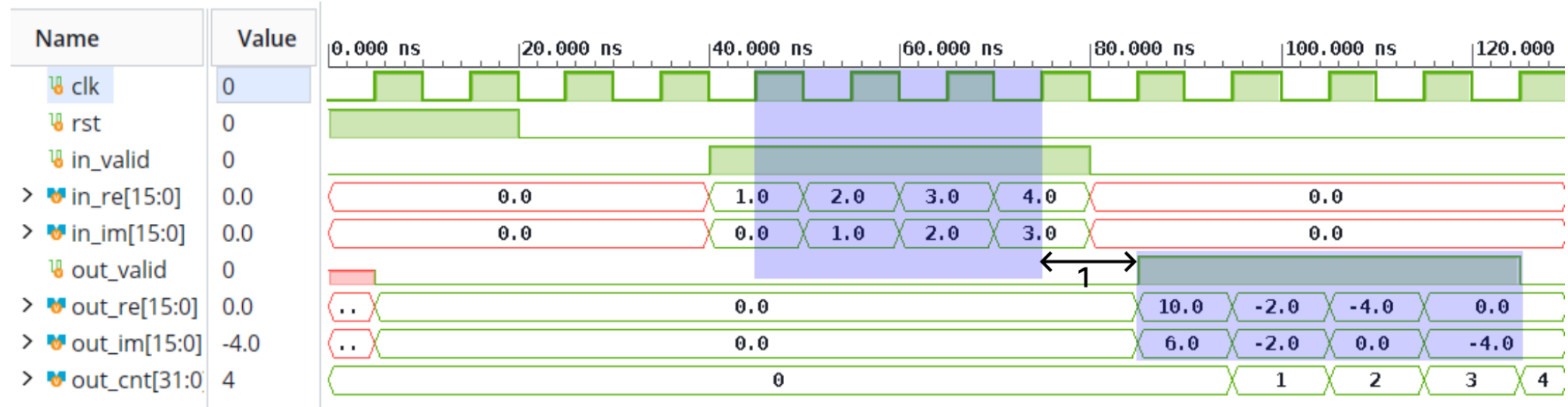


Figure 15: Simulation result for 4 point complex test signal in R2SDC

3.6 Winograd

- Actually a fast convolution algorithm for reducing multiplications
- Higher number of adders and higher routing complexity
- Particularly used in ASIC contexts and small sized DFTs in CT algorithm
- Implementation details vary for different lengths

3.6.1 Derivation for 5-point

5 point DFT is given by:

$$Y_k = \sum_{n=0}^4 x_n W^{nk}$$

Idea 1: prime length DFT can be done as circular convolution,

3.6 Winograd

$$\begin{bmatrix} Y(0) \\ Y(1) \\ Y(2) \\ Y(3) \\ Y(4) \end{bmatrix} = \begin{matrix} & \begin{matrix} 0 & 1 & 2 & 3 & 4 \end{matrix} \\ \begin{matrix} 0 \\ 1 \\ 2 \\ 3 \\ 4 \end{matrix} & \begin{bmatrix} W^0 & W^0 & W^0 & W^0 & W^0 \\ W^0 & W^1 & W^2 & W^3 & W^4 \\ W^0 & W^2 & W^4 & W^1 & W^3 \\ W^0 & W^3 & W^1 & W^4 & W^2 \\ W^0 & W^4 & W^3 & W^2 & W^1 \end{bmatrix} \end{matrix} \begin{bmatrix} x_0 \\ x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}$$

The first row corresponds to DC sum, and the rest of elements can be written as:

$$Y_k = x_0 + Y'(k), \quad k > 0$$

3.6 Winograd

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The first row corresponds to DC sum, and the rest of elements can be written as:

$$Y_k = x_0 + Y'(k), \quad k > 0$$

Where $Y'(k)$ is

$$\begin{bmatrix} Y'(1) \\ Y'(2) \\ Y'(3) \\ Y'(4) \end{bmatrix} = \begin{bmatrix} W^1 & W^2 & W^3 & W^4 \\ W^2 & W^4 & W^1 & W^3 \\ W^3 & W^1 & W^4 & W^2 \\ W^4 & W^3 & W^2 & W^1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}$$

Matrix can be rearranged to Toeplitz form and Winograd convolution algorithm can be applied.

3.6 Winograd

Idea 2: Rader Mapping When N is a prime number, set of non-zero numbers $\{1, 2, \dots, N - 1\}$ can be permuted using following scheme: $n = g^{-q} \pmod{N}$

$$Y'(g^p) = \sum_{q=0}^{N-2} x(g^{-q}) W^{g^{p-q}}$$

3.6 Winograd

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$$Y'(g^p) = \sum_{q=0}^{N-2} x(g^{-q}) W^{g^{p-q}}$$

Corresponding matrix form is:

$$\begin{bmatrix} Y'(1) \\ Y'(2) \\ Y'(4) \\ Y'(3) \end{bmatrix} = \begin{bmatrix} W^1 & W^3 & W^4 & W^2 \\ W^2 & W^1 & W^3 & W^4 \\ W^4 & W^2 & W^1 & W^3 \\ W^3 & W^4 & W^2 & W^1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_3 \\ x_4 \\ x_2 \end{bmatrix}$$

3.6 Winograd

Now the operation has become a length-4 cyclic convolution of sequences

$$a_q = x(2^{-q}) = \{x_0, x_1, x_2, x_3, x_4\}$$

$$b_{p-q} = W^{2^{p-q}} = \{W^1, W^2, W^4, W^3\}$$

$$Y'(2^p) = \sum_{q=0}^3 a_q \cdot b_{(p-q) \pmod{4}}$$

3.6 Winograd

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$$Y'(2^p) = \sum_{q=0}^3 a_q \cdot b_{(p-q) \pmod{4}}$$

Expressing this sequence as polynomials:

$$ax'(z) = x_1 + x_3z + x_4z^2 + x_2z^3 \quad W'(z) = W^1 + W^2z + W^4z^2 + W^3z^3$$

The circular convolution is given by: $Y'(z) = x'(z)W'(z) \pmod{z^4 - 1}$.

The required values are coefficients of $Y'(z)$

3.6 Winograd

Winograd algorithm:

1. Choose a cyclotomic polynomial $m(z)$ with $\deg(m) > \deg[W'x']$ and factor it into $k + 1$ relatively prime polynomials with real coefficients: $m^{(0)}, m^{(1)}, \dots, m^{(k)}$
2. Let $M^{(i)} = m/m^{(i)}$. Use Euclidean GCD algorithm to solve Bezout's identity and find $N^{(i)}$:

$$m^{(i)}n^{(i)} + M^{(i)}N^{(i)} = 1$$

3. For $i = 0, 1, \dots, k$.compute:

$$w^{(i)} = W'(z) \bmod m^{(i)} \quad x^{(i)} = x'(z) \bmod m^{(i)} \quad y^{(i)} = w^{(i)}x^{(i)} \bmod m^{(i)}$$

4. Compute $Y'(z)$ using:

$$Y'(z) = \left[\sum_{i=0}^k y^{(i)} N^{(i)} M^{(i)} \right] \bmod m(z)$$

3.6 Winograd

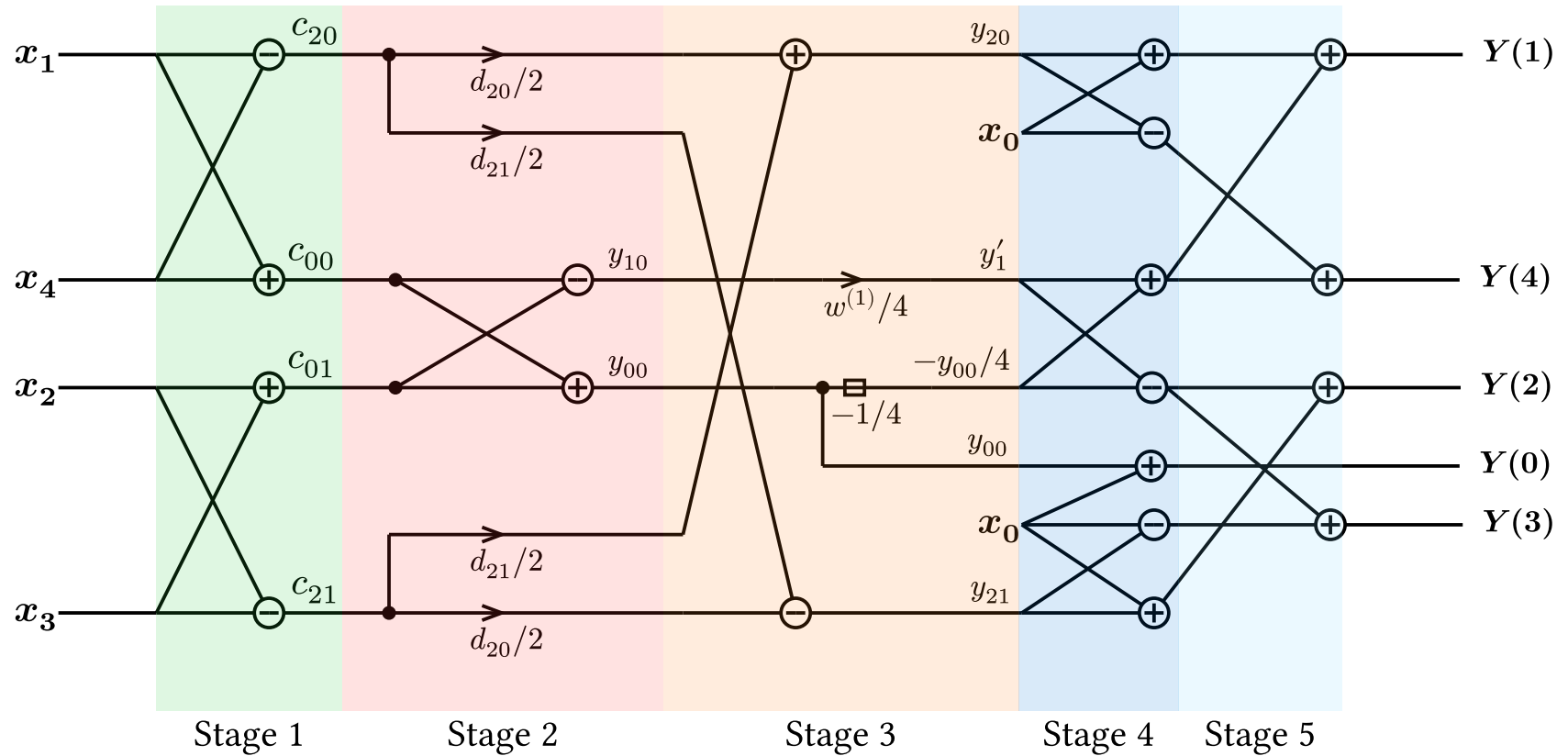


Figure 16: Signal flow of modified Winograd algorithm for 5 point DFT.

3.6 Winograd

3.6.2 16 Point Winograd algorithm

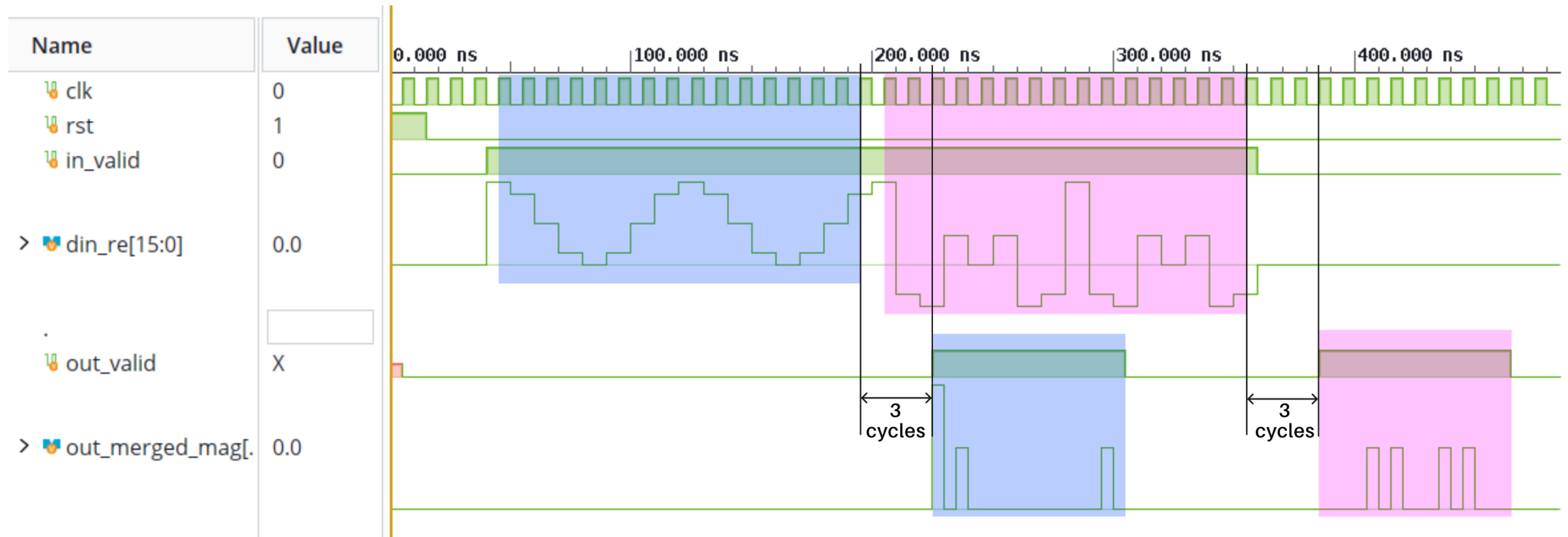


Figure 17: Output of 16 point Winograd implementation for 2 consecutive inputs.

4. Comparison

4.1 Comparison of 16 point transforms

Techniques	Total LUT	Flip-Flops	DSPs	Power (mW)	$F_{\max}$ (MHz)	Throughput (MSa/s)	Latency (cycles)
In-place	1611	555	4	104	79.2	26.41	49
R2MDC	419	645	12	187	137.1	274.2	10
R2SDC	516	327	12	154	118.3	118.3	17
Winograd	2330	2850	20	208	136.4	136.4	19

Table 1: Comparison of various 16 point techniques

- R2MDC seems to be high performing algorithm
- However it and SDC needs buffer and reorderer in subsequent sections

4.2 Comparison of scalable techniques

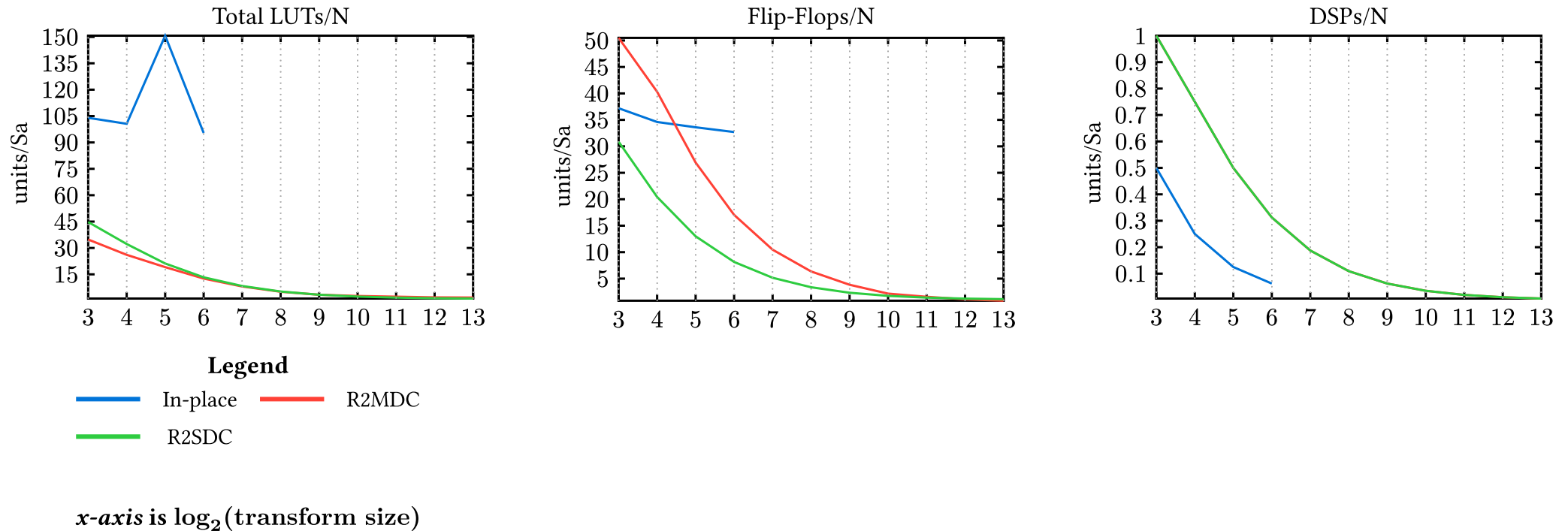


Figure 18: Resource and performance analysis of scalable techniques.

4.2 Comparison of scalable techniques

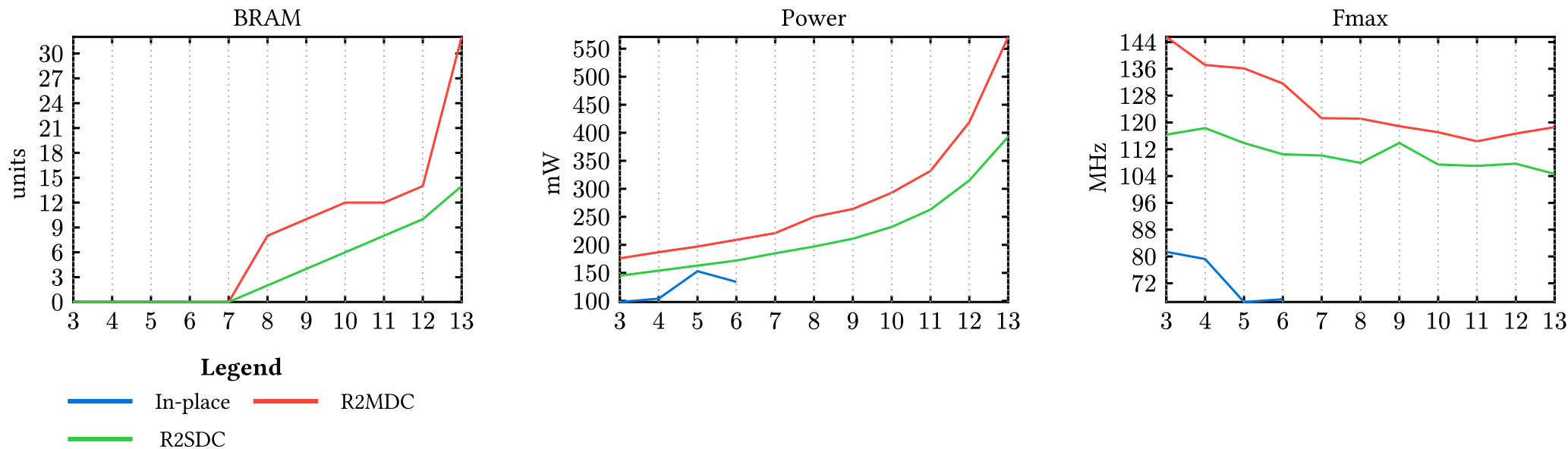


Figure 19: Resource and performance analysis of scalable techniques.

4.2 Comparison of scalable techniques

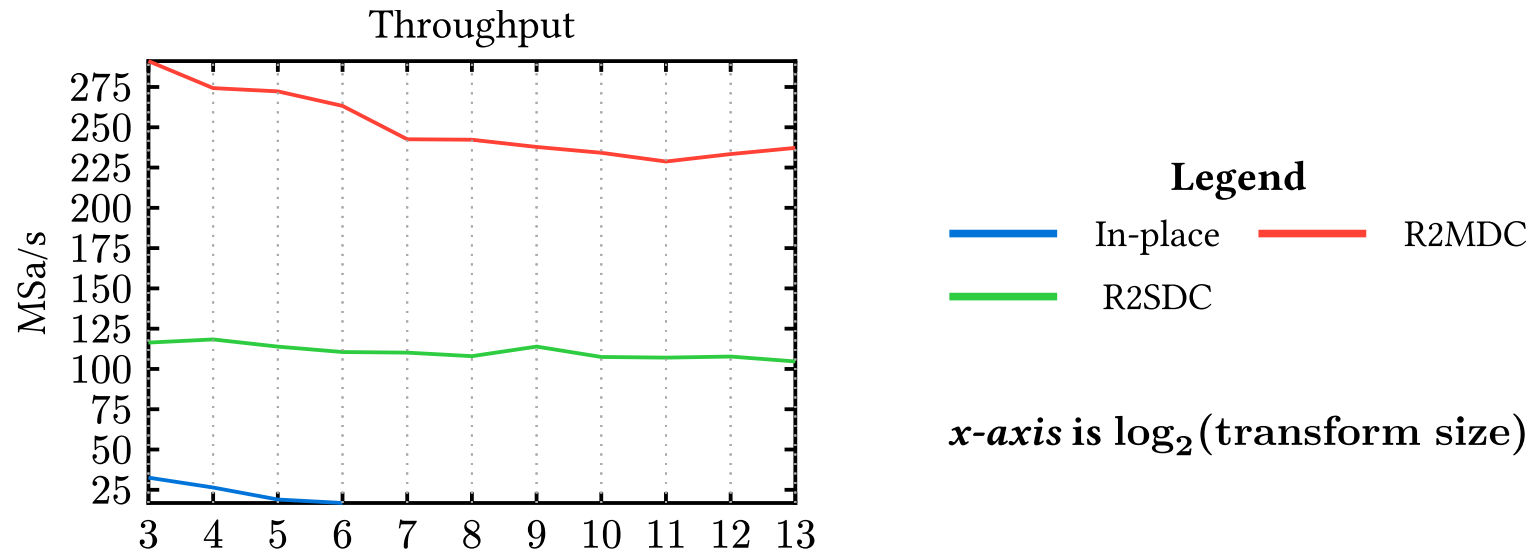


Figure 20: Resource and performance analysis of scalable techniques.

5. Conclusion

5. Conclusion

- Evaluated FFT techniques and their resource utilization on an Artix-7 FPGA.
- Choice of technique depends heavily on specific performance (latency, throughput, etc.), hardware constraints, etc.
- **In-place Radix-2 DIT**: Burst architecture, For low power, low speed systems.
- **R2MDC**: for high-throughput systems in which input data is available ahead of time. Outputs in bit-reversed order.
- **R2SDC**: Burst architecture. uses roughly half the FF of MDC. The outputs are in bit-reversed order.
- **Winograd**: Used in cases where hardware multipliers are scarce. Not well suited for FPGA

6. Future scope

6. Future scope

- Use of built in BRAM for reducing number of FF (codebase will not be tool/platform agnostic)
- Optimizing existing implementation
- Further routing can be reduced by considering case of real valued signals.

Thank you